

Solutions of Exam 2

M. Siddikov

July 4, 2024

Problem 1 - Exam 2

- ① (7 points) A pointlike nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric harmonic potential

$$U(r, t) = \frac{k(t) r^2}{2}, \quad k(t) = \frac{\kappa}{t^2}, \quad \kappa > 0. \quad (1)$$

- ① (2 points) Identify all continuous symmetries of the action. Analyze if there are symmetries of type

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \beta t, \quad \alpha, \beta = \text{const}$$

which do not change the equations of motion.

- ② (3 points) Construct the integrals of motion which might exist according to Noether's theorem. Demonstrate that these quantities are conserved using the equations of motion.
- ③ (2 points) We have seen in lectures that in static (time-independent) Coulomb potential the particle's trajectory always remains in the plane (so-called scattering plane). Analyze if this property remains valid in the time-dependent potential given in Eq. (1). Analyze if the Kepler's laws remain valid in this potential field.

Problem 1 (derivation of symmetry)

We can start with

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \beta t, \quad \alpha, \beta \in \mathbb{R}$$

The action transforms as

$$S \rightarrow S' = \int dt \left(\frac{\alpha^2}{\beta} \frac{m \dot{\mathbf{r}}^2}{2} - \frac{\alpha^2}{\beta} \frac{\kappa r^2}{2t^2} \right) = \frac{\alpha^2}{\beta} \int dt \left(\frac{m \dot{\mathbf{r}}^2}{2} - \frac{\kappa r^2}{2t^2} \right) = \frac{\alpha^2}{\beta} S$$

- We should NOT forget extra " β " which comes from $\int dt$
- The equations of motion remain invariant for any choice of α, β , however the action is invariant only if

$$\frac{\alpha^2}{\beta} = 1, \quad \alpha = \sqrt{\beta}$$

- Only in the latter case we can apply the Noether theorem and construct the integrals of motion associated with this symmetry.

Problem 1 (proof of invariance)

We analyzed symmetry

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \alpha^2 t, \quad \alpha \in \mathbb{R}$$

in Lecture 7, so we repeat almost literally those results. However, now energy $E \neq \text{const}$

$$S = \int dt \left(\frac{m \dot{\mathbf{r}}^2}{2} - \frac{\kappa r^2}{2t^2} \right)$$

Change of different quantities under rescaling:

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \alpha^2 t, \quad \alpha = e^\lambda \approx 1 + \lambda \quad (1)$$

$$\frac{m \dot{\mathbf{r}}^2}{2} \rightarrow \frac{1}{\alpha^2} \frac{m \dot{\mathbf{r}}^2}{2}, \quad U(r) \rightarrow \frac{1}{\alpha^2} U(r)$$

$$\Rightarrow S = \int dt L \rightarrow \int \cancel{\alpha^2} dt \frac{1}{\cancel{\alpha^2}} L = \text{const}$$

This proves that (1) is indeed a continuous symmetry of the action.

Problem 1 (derivation of invariant)

To find the associated Noether invariant, consider $\alpha = \alpha(t) = e^{\lambda(t)}$, $|\lambda| \ll 1$ and pick up $\mathcal{O}(\dot{\lambda})$ term in the expansion of the action:

$$d\mathbf{r} \rightarrow (d\mathbf{r} + \mathbf{r}d\lambda) = (d\mathbf{r} + \mathbf{r}\dot{\lambda}dt), \quad dt \rightarrow (1 + 2\dot{\lambda}t) dt$$

$$\mathbf{v} \equiv \frac{d\mathbf{r}}{dt} \rightarrow \frac{\mathbf{v} + \mathbf{r}\dot{\lambda}}{1 + 2\dot{\lambda}t} \approx \mathbf{v} + \dot{\lambda}(\mathbf{r} - 2t\mathbf{v})$$

$$\begin{aligned} S &\equiv \int dt L \rightarrow \int dt L (1 + 2\dot{\lambda}t) + \int dt m \mathbf{v} \cdot (\mathbf{r} - 2t\mathbf{v}) \dot{\lambda} \\ &= \int dt [m \mathbf{v} \cdot (\mathbf{r} - 2t\mathbf{v}) + 2tL] \dot{\lambda} \end{aligned}$$

Problem 1 - Exam 2

$$\begin{aligned} Q &= \frac{\partial \delta L}{\partial \dot{\lambda}} = m \mathbf{v} \cdot (\mathbf{r} - 2t \mathbf{v}) + 2tL = \\ &= m \mathbf{v} \cdot (\mathbf{r} - 2t \mathbf{v}) + 2t \left(\frac{m \mathbf{v}^2}{2} - U(r, t) \right) \\ &= \mathbf{p} \cdot \mathbf{r} - 2 \left(\frac{m \mathbf{v}^2}{2} + U(r, t) \right) t \end{aligned}$$

– Check that $Q = \text{const}$ using equations of motion and explicit expression for potential:

$$\begin{aligned} \frac{dQ}{dt} &= \dot{\mathbf{p}} \cdot \mathbf{r} + \mathbf{p} \cdot \dot{\mathbf{v}} - 2 \left(\frac{m \mathbf{v}^2}{2} + U(r, t) \right) - 2t \left(\frac{\mathbf{p} \cdot \dot{\mathbf{p}}}{m} + \frac{\partial U(r, t)}{\partial t} + \mathbf{v} \cdot \frac{\partial U(r, t)}{\partial \mathbf{r}} \right) \\ &= -\frac{\partial U}{\partial \mathbf{r}} \cdot \mathbf{r} + \cancel{\mathbf{p} \cdot \dot{\mathbf{v}}} - 2 \left(\cancel{\frac{m \mathbf{v}^2}{2}} + U(r, t) \right) - \\ &\quad - 2t \left(-\frac{1}{m} \cancel{\mathbf{p} \cdot \frac{\partial U(r, t)}{\partial \mathbf{r}}} + \frac{\partial U(r, t)}{\partial t} + \cancel{\mathbf{v} \cdot \frac{\partial U(r, t)}{\partial \mathbf{r}}} \right) = \left[-\mathbf{r} \cdot \frac{\partial}{\partial \mathbf{r}} - 2 - 2t \frac{\partial}{\partial t} \right] U \\ &= [-2 - 2 - 2 \cdot (-2)] U(r, t) = 0 \end{aligned}$$

as expected (!!!)

Problem 1 - Exam 2

- 1 Since $\mathbf{M} = \text{const}$, repeating discussion from Lecture 7, conclude that trajectory always remains in plane $\perp$ to vector $\mathbf{M}$.
- 2 Since $|\mathbf{M}| = \text{const}$, repeating discussion from Lecture 7, conclude that the 2nd law of Kepler remains valid
- 3 The first and the third laws of Kepler are NOT valid, since they use explicit trajectory for their proof.

Problem 2 - Exam 2

(8 points) A relativistic particle of mass m moves in the spherically symmetric potential $U(r) = \alpha/r + \beta/r^2$.

- Write out the lagrangian of the system, and the corresponding equations of motion
- Identify all the symmetries of the action and construct all integrals of motion which might exist in this problem.
- Try to integrate the equations of motion and find trajectories for bound orbits.
- Analyze the behavior of your solution in the limit $c \rightarrow \infty$. Assuming that c is large yet finite, find explicitly the first relativistic correction to the trajectory.

Problem 2 - general relativistic case

–Lagrangian:

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - U(r) = -mc^2 \sqrt{1 - \frac{\dot{r}^2 + r^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)}{c^2}} - \alpha/r - \beta/r^2$$

–generalized coordinates: r, θ, ϕ

–generalized momenta:

$$p_r = \frac{\partial L}{\partial \dot{r}} = \gamma m \dot{r}, \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = \gamma m r^2 \dot{\theta}, \quad p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \gamma m r^2 \sin^2 \theta \dot{\phi}, \quad \gamma \equiv \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

–Explicit Euler-Lagrange equations:

$$\frac{d(\gamma m \dot{r})}{dt} = -\kappa r + 2\gamma m r (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2),$$

$$\frac{d(\gamma m r^2 \dot{\theta})}{dt} = \frac{\gamma m r^2 \sin 2\theta \dot{\phi}^2}{2}, \quad \frac{d(\gamma m r^2 \sin^2 \theta \dot{\phi})}{dt} = 0$$

–Last equation: $p_\phi = \text{const}$, can use to eliminate $\dot{\phi}$:

$$\frac{d(\gamma m \dot{r})}{dt} = -\kappa r + 2\gamma m r \left(\dot{\theta}^2 + \frac{p_\phi^2}{\gamma^2 m^2 r^4 \sin^2 \theta} \right), \quad \frac{d(\gamma m r^2 \dot{\theta})}{dt} = \frac{p_\phi^2 \cos \theta}{\gamma m r^2 \sin^3 \theta}.$$

Problem 2 - Exam 2

–Lagrangian:

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - U(r) = -mc^2 \sqrt{1 - \frac{\dot{r}^2 + r^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)}{c^2}} - \alpha/r - \beta/r^2$$

- Explicit Euler-Lagrange equations are extremely complicated, so will use integrals of motion instead to find the trajectory
- Symmetries** & integrals of motion:

$$\phi \rightarrow \phi + \phi_0 \Rightarrow p_\phi \equiv M_z = \text{const}$$

$$t \rightarrow t + t_0 \Rightarrow E = \text{const}$$

$$E = p_r \cdot \dot{r} + p_\theta \cdot \dot{\theta} + p_\phi \cdot \dot{\phi} - L =$$

$$= mc^2 \gamma + U(r), \quad \kappa > 0$$

–In spherical coordinates we don't see other symmetries, but ...

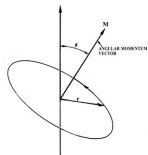
Problem 2 - Exam 2

- In Cartesian coordinates we know that potential is invariant w.r.t. rotation around *any* axis



- All components of $\vec{M}$ are conserved, $\vec{M}$ keeps its direction
- In spherical coordinates the orientation of basis vectors $\mathbf{e}_\theta, \mathbf{e}_\phi$ changes, so we don't see this property (actually there is a hidden symmetry which mixes θ, ϕ in non-trivial way)

Consequences of $\vec{M} = \text{const}$:



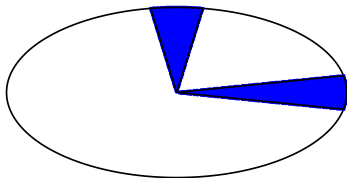
- Particle always moves in a plane orthogonal to $\vec{M}$, i.e. vectors $(\vec{r}(t), \vec{v}(t))$ always remain in this plane, for $\forall t$.

- We may choose plane $\theta = \pi/2$ as a plane of motion, like in nonrelativistic case

$$M \equiv \gamma m r^2 \dot{\phi} = \frac{m r^2 \dot{\phi}}{\sqrt{1 - v^2/c^2}} = \text{const}$$

Second law of Kepler is NOT valid

- Due to $\gamma \neq 1$, see that $r^2 d\phi \neq \text{const}$ for relativistic particle in general.
- It works ONLY for special case when particle has $v \ll c$ or $v = \text{const}$ (energy conservation: this is possible only along circular orbit with $r = \text{const}$)



Problem 2 - Exam 2

- Energy conservation (motion plane $\theta = \pi/2$ orthogonal to $\mathbf{M}$)

$$E = m\gamma c^2 + U(r)$$

$$\gamma = \left[1 - \frac{\dot{r}^2 + r^2 \dot{\phi}^2}{c^2} \right]^{-1/2}$$

$$M_z = m\gamma r^2 \dot{\phi} = \text{const}$$

$$\Rightarrow M_z^2 \left(1 - \frac{\dot{r}^2 + r^2 \dot{\phi}^2}{c^2} \right) = m^2 r^4 \dot{\phi}^2$$

$$\dot{\phi} = \frac{M_z}{mr^2} \sqrt{\frac{1 - \dot{r}^2/c^2}{1 + M_z^2/(m^2 r^2 c^2)}}$$

$$\gamma = \frac{M_z}{mr^2 \dot{\phi}} = \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}}$$

$$E = mc^2 \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} + U(r) \quad (1)$$

In nonrelativistic case ($c \gg 1$) this result yields

$$E_{\text{nonrel}} = E - mc^2 = \quad (2)$$

$$m\dot{r}^2 + \frac{M_z^2}{mr^2} + U(r) + \mathcal{O}\left(\frac{1}{c^2}\right)$$

and we interpret the last 2 terms as an effective potential.

–If $\beta > 0$, the particle never reaches $r = 0$. in the nonrelativistic limit, since potential U goes to infinity there. If $\beta < 0$, the potential may catch particles with small angular momenta $M_z^2/2m < \beta$.

Problem 2 - Exam 2

$$E = mc^2 \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} + U(r) \quad (1)$$

– We'll try to rewrite (1) as:

$$\frac{mc^2}{2} = \frac{m\dot{r}^2}{2} + \frac{m^3 c^6}{2} \left(1 + \left(\frac{M_z}{mrc} \right)^2 \right) / (E - U(r))^2 \quad (2)$$

Can interpret the last term as “potential” and use it for qualitative interpretation, however there is some peculiarity:

- * “Potential” depends on true energy E
- * Should consider mc^2 as “energy” (see left-hand side)
- * Large- c expansion near $E \approx mc^2$

$$V_{\text{eff}} \approx U(r) + \frac{M_z^2}{2mr^2} + \frac{(2M_z^2 + 3mr^2 U(r)) U(r)}{2m^2 r^2 c^2}$$

Problem 2 - Exam 2

Solution of EOMs:

$$\frac{dr}{dt} = f(r, E) = \sqrt{c^2 - m^2 c^6 (E - U(r))^{-2} \left(1 + \left(\frac{M_z}{mrc} \right)^2 \right)}$$

$$\int^r \frac{dr}{f(r, E)} = t - t_0 \quad (3)$$

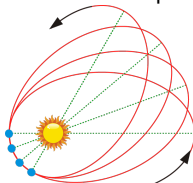
The integral over r is expressed in terms of elliptic functions.

–Equation for trajectory $r(\phi)$:

$$\frac{d\phi}{dr} = \frac{\dot{\phi}}{\dot{r}} = \frac{M_z}{mc^2 f(r, E)} (E - U(r)) \equiv g(r, E)$$

$$\Rightarrow \phi - \phi_0 = \int dr g(r, E)$$

*Trajectories are NOT ellipses, NOT even closed.



Corrections to NR trajectory

–We already got trajectory in general case.

$$t - t_0 = \int \frac{dr}{\sqrt{c^2 - m^2 c^6 (E - U(r))^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}} \quad (1)$$

$$\phi - \phi_0 = \int dr \frac{M_z (E - U(r))}{mc^2 \sqrt{c^2 - m^2 c^6 (E - U(r))^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}} \quad (2)$$

Can make expansion over $1/c$ to get NR limit, take into account $E = E_{\text{nonrel}} + mc^2$

Corrections to NR trajectory

$$t - t_0 \approx \int \frac{dr}{\sqrt{\frac{2}{m} \left(E_{\text{nonrel}} - U(r) - \frac{M_z^2}{2mr^2} \right)}} - \frac{\left(\frac{M_z^2}{mr^2} + \frac{3}{2} (U(r) - E_{\text{nonrel}}) \right) (U(r) - E_{\text{nonrel}})}{2c^2 \left(\frac{M_z^2}{2mr^2} + (U(r) - E_{\text{nonrel}}) \right) \sqrt{\frac{2}{m} \left(E_{\text{nonrel}} - U(r) - \frac{M_z^2}{2mr^2} \right)}} \quad (1)$$

$$\phi - \phi_0 \approx \int \frac{dr}{\sqrt{\frac{2}{m} \left(E_{\text{nonrel}} - U(r) - \frac{M_z^2}{2mr^2} \right)}} - \frac{M_z \left(\frac{2M_z^2}{mr^2} + \frac{7}{2} (U(r) - E_{\text{nonrel}}) \right) (U(r) - E_{\text{nonrel}})}{2c^2 \left(\frac{M_z^2}{2mr^2} + (U(r) - E_{\text{nonrel}}) \right) \sqrt{2m \left(E_{\text{nonrel}} - U(r) - \frac{M_z^2}{2mr^2} \right)}} \quad (2)$$

- The first terms in right-hand side of (6, 7) coincide with known non-relativistic results
- *Mathematica* fails to evaluate integrals in previous slide analytically, but after integration of red-colored terms in (1,2) gives result in terms of elementary functions (though very lengthy). These are desired relativistic corrections.